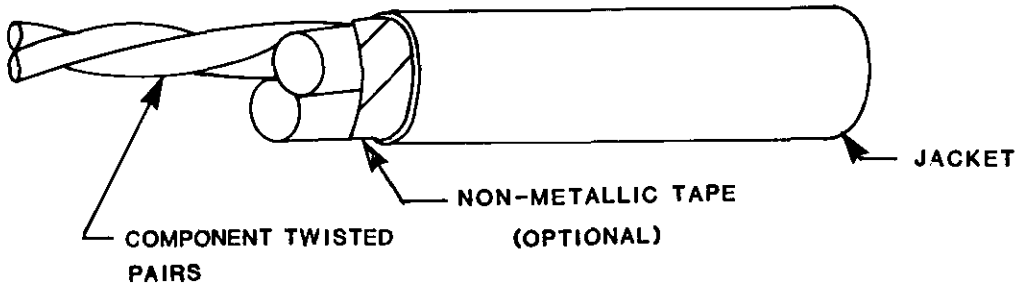
MIL-C-0024640/24A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/24
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, AWG 22, TYPE TTXW (INCLUDING VARIATION TYPES TTXOW AND TTXWA)

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24640. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24640.



SH 12771

(A) COMPONENT PAIRS:

First - Copper conductor, AWG 22, 7 stranding. Nominal diameter of stranded conductor 0.029 inch.
Second - MICA-glass tape, insulation 0.0029 inch (50 percent overlap maximum).
Third - Polyimide tape 0.001 inch (50 percent overlap minimum).
Fourth - Polyimide tape 0.001 inch with 0.0001 inch fluorinated ethylene propylene (FEP) both sides (50 percent overlap minimum).
Fifth - Modified aromatic polyimide RESIN coating 0.0005 inch minimum (0.001 inch nominal). Finished wire diameter 0.051 inch minimum and 0.055 inch maximum.

FIGURE 1. Construction (watertight with circuit integrity).

(A) denotes changes.

FSC 6145

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(A) TABLE I. Construction details - unarmored.

Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
TTXW-1-1/2	1-1/2	0.028	0.181	0.195	26.0
TTXW-3	3	.038	.285	.307	62.3
TTXW-5	5	.038	.331	.357	91.0
TTXW-10	10	.041	.456	.492	168.0
TTXW-15	15	.044	.527	.569	231.0
TTXW-20	20	.044	.577	.621	283.0
TTXW-30	30	.044	.684	.738	407.0
TTXW-40	40	.047	.790	.852	526.0

(A) TABLE II. Construction details - armored.

Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
TTXWA-1-1/2	1-1/2	0.028	0.231	0.245	41.0
TTXWA-3	3	.038	.335	.357	84.3
TTXWA-5	5	.038	.381	.407	119.4
TTXWA-10	10	.041	.506	.542	209.3
TTXWA-15	15	.044	.577	.619	269.6
TTXWA-20	20	.044	.627	.671	328.2
TTXWA-30	30	.044	.734	.788	463.2
TTXWA-40	40	.047	.840	.902	592.6

(A) TABLE III. Construction details - overall shield.

Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
TTXOW-1-1/2	1-1/2	0.127	0.031	0.253	0.273	67.9
TTXOW-3	3	.216	.038	.333	.359	108.0
TTXOW-5	5	.263	.038	.376	.406	134.0
TTXOW-15	15	.449	.044	.556	.600	299.0
TTXOW-20	20	.510	.044	.614	.662	361.0
TTXOW-30	30	.610	.047	.717	.772	480.0
TTXOW-40	40	.720	.047	.823	.887	631.0

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REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

(A) Finished wire:

For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature.
Voltage rating: 600 volts root mean square (rms) at sea level.
Blocking: Oven temperature, 200 ± 2°C (392 ± 3.6°F); no sticking of conductors.
Color: As specified in MIL-C-24640.
Flammability: 3 seconds (maximum) after-flame.
3.0 inches (maximum) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms-1,000 feet, minimum insulation resistance after humidity exposure.
Identification of product: Required for sizes 22 and larger.
Impulse dielectric test: 100 percent test; impulse voltage at 8 kV.
Insulation resistance: 2,500 megohms-1,000 feet (minimum).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch (maximum) at 180°C.
Surface resistance: 5 megohms-inches (minimum), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F). Change in measurement, 0.031 inch (maximum).
Wet dielectric test: 2,500 volts rms.

When this cable is manufactured with armor, the following additional requirements shall apply:

(a) Braided metal armor and paint shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(a) An overall tin-coated copper braid shall be added.
(b) Braid coverage shall be not less than 85 percent determined in accordance with MIL-C-27500.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	16.52
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Conductor to shield (TTXOW only)	1000
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	2500
Conductor and shield continuity (type TTXOW only)	No failure
Jacket flaws (type TTXOW only)	No failure
(A) Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
(A) Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	No failure
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type TTXWA only) - conformance to material, construction and coverage	No failure
Shield (type TTXOW only) - conformance to material, construction and coverage	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance (type TTXOW only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	No failure
Finished wire	No failure

Preparing activity:
Navy - SH
(Project 6145-N322-24)

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